

Practice 4

Following the Practice 3:

Define a **Warrior** class with the attributes String Name, int Life, and int Magic, and the attack method void NewMoon(). The attack target can be a warrior, a witch, or a priest:

- Deduct 25 life points if the target is a warrior.
- Deduct 40 life points if the target is a witch.
- Deduct 30 life points if the target is a priest.

Each attack consumes 10 magic points. If magic points are less than 10, the attack is invalid, and a message is printed. If the attacked character's life points are ≤ 0 , print the name of the deceased character and a death message.

Initial life points are 400, and initial magic points are 100.

Define a **Witch** class with the attributes String Name, int Life, and int Magic, and the magical attack method void SmallFire(). The attack target can be a warrior, a witch, or a priest:

- Deduct 40 life points if the target is a warrior.
- Deduct 60 life points if the target is a witch.
- Deduct 50 life points if the target is a priest.

Each attack consumes 25 magic points. If magic points are less than 25, the spell is invalid, and a message is printed. If the attacked character's life points are ≤ 0 , print the name of the deceased character and a death message.

Initial life points are 280, and initial magic points are 200.

Define a **Priest** class with the attributes String Name, int Life, and int Magic, and the attack method void Dark(). The attack target can be a warrior, a witch, or a priest:

- Deduct 30 life points if the target is a warrior.
- Deduct 50 life points if the target is a witch.
- Deduct 40 life points if the target is a priest.

Each attack consumes 20 magic points. If magic points are less than 20, the attack is invalid, and a message is printed. If the attacked character's life points are ≤ 0 , print the name of the deceased character and a death message.

Initial life points are 340, and initial magic points are 160.

Define a parent class **ROLE** with the attributes String Name, int Life, and int Magic, and have the Warrior, Witch, and Priest classes inherit from it.

Using polymorphism, design the following attack methods:

- NewMoon(Role R) for the Warrior.
- SmallFire(Role R) for the Witch.
- Dark(Role R) for the Priest.